

Q 1 Find Laplace Transformations of following functions -

$$(i) f(t) = \begin{cases} \sin t & 0 \leq t < \pi \\ 0 & t \geq \pi \end{cases}$$

$$\mathcal{L}[f(t)] = \int_0^{\infty} e^{-st} f(t) dt.$$

$$= \int_0^{\pi} e^{-st} \sin t dt + \int_{\pi}^{\infty} e^{-st} (0) dt.$$

$$= \int_0^{\pi} e^{-st} \sin t dt.$$

$$\left[\int e^{at} \sin bt = \frac{e^{at}}{a^2 + b^2} [a \sin bt - b \cos bt] \right]$$

$$= \frac{e^{-st}}{s^2 + 1} [-s \sin t - \cos t] \Big|_0^{\pi}.$$

$$= \frac{1}{s^2 + 1} \left[e^{-s\pi} (0 - (-1)) - 1(-1) \right] = \boxed{\frac{1 + e^{-s\pi}}{s^2 + 1}} \underline{\underline{A}}$$

$$(ii) \mathcal{L}[\cos(at + b)]$$

$$\mathcal{L}[\cos at \cos b - \sin at \sin b]$$

$$[\cos(a+b) = \cos a \cos b - \sin a \sin b]$$

$$= \cos b \left[\frac{s}{s^2 + a^2} \right] - \sin b \left[\frac{a}{s^2 + a^2} \right]$$

$$= \boxed{\frac{s \cos b - a \sin b}{s^2 + a^2}} \underline{\underline{A}}$$

$$(iii) \mathcal{L} \left[\frac{\sin t \sin 5t}{t} \right]$$

$$[2 \sin a \sin b = \cos(a-b) - \cos(a+b)]$$

$$\frac{1}{2} \mathcal{L} \left[\frac{\cos 4t - \cos 6t}{t} \right]$$

$$\frac{1}{2} \mathcal{L}[\cos 4t] = \frac{s}{s^2+16}$$

$$\mathcal{L} \left[\frac{\cos 4t}{t} \right] = \frac{1}{2} \int_s^\infty \frac{2s}{s^2+16} ds$$

$$= \frac{1}{2} [\log(s^2+16)]_s^\infty$$

$$= \frac{1}{2} [\log \infty - \log(s^2+16)]$$

$$= -\frac{1}{2} \log(s^2+16)$$

$$\mathcal{L}[\cos 6t] = \frac{s}{s^2+36}$$

$$\mathcal{L} \left[\frac{\cos 6t}{t} \right] = \frac{1}{2} \int_s^\infty \frac{2s}{s^2+36} ds$$

$$= \frac{1}{2} [\log(s^2+36)]_s^\infty$$

$$= \frac{1}{2} [\log \infty - \log(s^2+36)]$$

$$= -\frac{1}{2} \log(s^2+36)$$

$$\frac{1}{2} \left[-\frac{1}{2} \log(s^2+16) + \frac{1}{2} \log(s^2+36) \right]$$

$$\frac{1}{4} [\log(s^2+36) - \log(s^2+16)] = \frac{1}{4} \log \left(\frac{s^2+36}{s^2+16} \right)$$

$$(iv) \mathcal{L} \left[\frac{1 - \cos 3t}{t} \right]$$

$$\mathcal{L}[1] = \frac{1}{s}$$

$$\mathcal{L} \left[\frac{1}{t} \right] = \int_s^\infty \frac{1}{s} ds$$

$$= [\log(s)]_s^\infty = \log \infty - \log s$$

$$= -\log s$$

$$\mathcal{L}[\cos 3t] = \frac{s}{s^2+9}$$

$$\mathcal{L} \left[\frac{\cos 3t}{t} \right] = \frac{1}{2} \int_s^\infty \frac{2s}{s^2+9} ds$$

$$= \frac{1}{2} [\log(s^2+9)]_s^\infty$$

$$= \frac{1}{2} [\log \infty - \log(s^2+9)]$$

$$= -\frac{1}{2} \log(s^2+9)$$

$$\therefore \mathcal{L}\left[\frac{1}{t} - \frac{\cos 3t}{t}\right] = -\log s + \frac{1}{2} \log(s^2+9)$$

$$= \frac{1}{2} [\log(s^2+9) - 2\log s]$$

$$= \frac{1}{2} [\log(s^2+9) - \log s^2]$$

$$= \frac{1}{2} \log \frac{s^2+9}{s^2} \quad \underline{\underline{A}}$$

$$(V) \mathcal{L}\left[\int_0^t \frac{e^t \sin t}{t} dt\right]$$

$$\mathcal{L}[\sin t] = \frac{1}{s^2+1}$$

$$\mathcal{L}[e^t \sin t] = \frac{1}{(s-1)^2+1} \quad (\text{using 1st shifting theorem})$$

$$\mathcal{L}\left[\frac{e^t \sin t}{t}\right] = \int_s^\infty \frac{1}{(x-1)^2+1} dx$$

$$= \tan^{-1}(s-1) \Big|_s^\infty$$

$$= \tan^{-1} \infty - \tan^{-1}(s-1)$$

$$= \frac{\pi}{2} - \tan^{-1}(s-1)$$

$$\mathcal{L}\left[\frac{e^t \sin t}{t}\right] = \boxed{\cot^{-1}(s-1)}$$

$$\left[\int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} \right]$$

$$\left[\cot^{-1} x + \tan^{-1} x = \frac{\pi}{2} \right]$$

$$\mathcal{L}\left[\int_0^t \frac{e^t \sin t}{t} dt\right] = \boxed{\frac{1}{s} \cot^{-1}(s-1)} \quad \underline{\underline{A}}$$

$$(vi) \mathcal{L}\left[1 + 2\sqrt{t} + \frac{3}{\sqrt{t}}\right]$$

$$\mathcal{L}[1] + 2\mathcal{L}[t^{1/2}] + 3\mathcal{L}[t^{-1/2}]$$

$$= \frac{1}{s} + 2 \frac{\frac{1}{2} \sqrt{\frac{1}{2}}}{s^{3/2}} + 3 \frac{\sqrt{-\frac{1}{2}+1}}{s^{1/2}}$$

$$= \boxed{\frac{1}{s} + \frac{\sqrt{\pi}}{s^{3/2}} + 3 \frac{\sqrt{\pi}}{s}} \underline{\underline{\Delta}}$$

$$\left[\begin{aligned} \mathcal{L}[t^n] &= \frac{\Gamma(n+1)}{s^{n+1}} \\ &= \frac{n! \sqrt{\pi}}{s^{n+1}} \end{aligned} \right]$$

$$[\Gamma(n+1) = n! \sqrt{\pi}]$$

$$(vii) \mathcal{L}[\sin \sqrt{t}]$$

$$\sin t = t - \frac{t^3}{3!} + \frac{t^5}{5!} - \frac{t^7}{7!} + \dots$$

$$\sin \sqrt{t} = t^{1/2} - \frac{t^{3/2}}{3!} + \frac{t^{5/2}}{5!} - \dots$$

$$\mathcal{L}[\sin \sqrt{t}] = \mathcal{L}[t^{1/2}] - \frac{1}{6} \mathcal{L}[t^{3/2}] + \frac{1}{120} \mathcal{L}[t^{5/2}] - \dots$$

$$= \frac{\sqrt{\frac{1}{2}+1}}{s^{1/2+1}} - \frac{1}{6} \frac{\sqrt{\frac{3}{2}+1}}{s^{3/2+1}} + \frac{1}{120} \frac{\sqrt{\frac{5}{2}+1}}{s^{5/2+1}} - \dots$$

$$= \frac{\frac{1}{2} \sqrt{\frac{1}{2}}}{s^{3/2}} - \frac{1}{6} \frac{\frac{3}{2} \sqrt{\frac{3}{2}}}{s^{5/2}} + \frac{1}{120} \frac{\frac{5}{2} \sqrt{\frac{5}{2}}}{s^{7/2}} - \dots$$

$$= \frac{1}{2} \frac{\sqrt{\pi}}{s^{3/2}} - \frac{1}{4} \cdot \frac{1}{2} \frac{\sqrt{\pi}}{s^{5/2}} + \frac{1}{120} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \frac{\sqrt{\pi}}{s^{7/2}} - \dots$$

$$= \frac{1}{2} \frac{\sqrt{\pi}}{s^{3/2}} - \frac{1}{8} \frac{\sqrt{\pi}}{s^{5/2}} + \frac{1}{64} \frac{\sqrt{\pi}}{s^{7/2}} - \dots$$

$$= \frac{\sqrt{\pi}}{2s^{3/2}} \left[1 - \frac{1}{4s} + \frac{1}{32s^2} - \dots \right]$$

$$= \frac{\sqrt{\pi}}{2s^{3/2}} \left[1 - \frac{1}{4s} + \frac{1}{2!(4s)^2} - \dots \right] = \boxed{\frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}} \underline{\underline{\Delta}}$$

$$(9) \int_0^{\infty} t^3 e^{-t} \sin t \, dt$$

$$\mathcal{L}[\sin t] = \frac{1}{s^2+1}$$

$$\mathcal{L}[t^3 \sin t] = \frac{d^3}{ds^3} \left[\frac{1}{s^2+1} \right]$$

$$= \frac{d^2}{ds^2} \left[\frac{-2s}{(s^2+1)^2} \right] = -2 \frac{d}{ds} \left[\frac{s}{(s^2+1)^2} \right]$$

$$= -2 \frac{d}{ds} \left[\frac{(s^2+1)^2 - s \cdot 4s(s^2+1)}{(s^2+1)^4} \right] = -2 \frac{d}{ds} \left[\frac{s^2+1-4s^2}{(s^2+1)^3} \right]$$

$$= -2 \frac{d}{ds} \left[\frac{1-3s^2}{(s^2+1)^3} \right]$$

$$= -2 \left[\frac{-6(s^2+1)^3 s - (1-3s^2)6s(s^2+1)^2}{((s^2+1)^3)^2} \right]$$

$$= -2 \left[\frac{-6s(s^2+1)^3 - 6s(1-3s^2)(s^2+1)^2}{(s^2+1)^6} \right]$$

$$\mathcal{L}[t^3 \sin t]$$

$$\mathcal{L}[f(t)] = 12 \left[\frac{s(s^2+1) + s(1-3s^2)}{(s^2+1)^4} \right]$$

$$\int_0^{\infty} e^{-st} \frac{t^3 \sin t}{f(t)} \, dt = 12 \left[\frac{s(s^2+1) + s(1-3s^2)}{(s^2+1)^4} \right]$$

$$s = 1$$

$$\int_0^{\infty} e^{-t} t^3 \sin t \, dt = 12 \left[\frac{2-2}{1} \right] = \boxed{0} \underline{\underline{A}}$$

(3) Prove (i) $\int_0^{\infty} \frac{e^{-at} - e^{-bt}}{t} dt = \log \frac{b}{a}$

$\mathcal{L}\left[\frac{1}{t}\right] = ? \quad \mathcal{L}[1] = \frac{1}{s} \quad \rightarrow \int_0^{\infty} \frac{e^{-as}}{t} dt = \int_0^{\infty} \frac{e^{-bs}}{t} dt \quad \text{--- (1)}$

$\mathcal{L}\left[\frac{1}{t}\right] = \int_s^{\infty} \frac{1}{s} ds = [\log s]_s^{\infty} = \log \infty - \log s$

$\mathcal{L}[f(t)] = \boxed{-\log s}$

$\int_0^{\infty} e^{-st} \frac{1}{t} dt = -\log s \quad \left| \quad \int_0^{\infty} e^{-bt} \frac{1}{t} dt = -\log b \right.$
 $(s=a) \quad s=b$

$\int_0^{\infty} \frac{e^{-at}}{t} dt = \boxed{-\log a} \quad \left| \quad \int_0^{\infty} \frac{e^{-bt}}{t} dt = \boxed{-\log b} \right.$

Put values in (1).

$-\log a + \log b = \boxed{\log \frac{b}{a}} = \text{RHS}$
Hence proved

(ii) $\int_0^{\infty} \frac{e^{-t} \sin^2 t}{t} dt = \frac{1}{4} \log 5$

$\mathcal{L}[\sin^2 t] = \frac{1}{2} \mathcal{L}[1 - \cos 2t]$

$\mathcal{L}[\sin^2 t] = \frac{1}{2} \left[\frac{1}{s} - \frac{s}{s^2 + 4} \right]$

$\mathcal{L}\left[\frac{\sin^2 t}{t}\right] = \frac{1}{2} \int_s^{\infty} \left[\frac{1}{s} - \frac{s}{s^2 + 4} \right] ds$

$= \frac{1}{2} \left[\log s - \frac{1}{2} \log(s^2 + 4) \right]_s^{\infty}$

$= \frac{1}{4} \left[2 \log s - \log(s^2 + 4) \right]_s^{\infty}$

$= \frac{1}{4} \left[\log \frac{s^2}{s^2 + 4} \right]_s^{\infty}$

$= \frac{1}{4} \left(\log \infty - \log \frac{s^2}{s^2 + 4} \right)$

$\mathcal{L}[f(t)]$

$\mathcal{L}\left[\frac{\sin^2 t}{t}\right] = \boxed{\frac{1}{4} \log \frac{s^2 + 4}{s^2}} \rightarrow$

$\int_0^{\infty} e^{-st} f(t) dt = \frac{1}{4} \log \frac{s^2 + 4}{s}$
 $(s=1)$

$\int_0^{\infty} \frac{e^{-t} \sin^2 t}{t} dt = \boxed{\frac{1}{4} \log 5} = \text{RHS}$

Hence proved

④ First Shifting Theorem.

Statement:- If $\mathcal{L}[f(t)] = F(s)$, then

$$\mathcal{L}[e^{at} f(t)] = F(s-a)$$

Proof:- LHS $\mathcal{L}[e^{at} f(t)] = \int_0^{\infty} e^{-st} e^{at} f(t) dt$

$$= \int_0^{\infty} e^{-(s-a)t} f(t) dt \quad s > a.$$
$$s-a = \lambda.$$

$$= \int_0^{\infty} e^{-\lambda t} f(t) dt$$

$$= \mathcal{L}[f(t)] = F(\lambda)$$

$$= F(s-a) = \text{RHS}$$

Hence proved

⑤ $\mathcal{L}\left[\int_0^t \int_0^t \int_0^t t \sin t \, dt \, dt \, dt\right]$

$$\mathcal{L}[\sin t] = \frac{1}{s^2+1}$$

$$\mathcal{L}[t \sin t] = \frac{d}{ds} \left[\frac{1}{s^2+1} \right]$$

$$\mathcal{L}[t \sin t] = \frac{-2s}{(s^2+1)^2}$$

$$\mathcal{L}\left[\int_0^t t \sin t \, dt\right] = \frac{-2s'}{s(s^2+1)^2} = \frac{-2}{(s^2+1)^2}$$

$$\mathcal{L}\left[\int_0^t \int_0^t t \sin t \, dt \, dt\right] = \frac{-2}{s(s^2+1)^2}$$

$$\mathcal{L}\left[\int_0^t \int_0^t \int_0^t t \sin t \, dt \, dt \, dt\right] = \boxed{\frac{-2}{s^2(s^2+1)^2}} \underline{\underline{Ans}}$$

(6) Find inverse Laplace transforms of following-

(i) $\mathcal{L}^{-1} \left[\frac{s+2}{s^2-4s+13} \right]$

$$\mathcal{L}^{-1} \left[\frac{s+2}{s^2-2(s)(2)+2^2-2^2+13} \right]$$

$$\mathcal{L}^{-1} \left[\frac{s+2}{(s-2)^2+9} \right]$$

$$\mathcal{L}^{-1} \left[\frac{s-2+4}{(s-2)^2+9} \right]$$

$$\mathcal{L}^{-1} \left[\frac{s-2}{(s-2)^2+3^2} \right] + 4 \mathcal{L}^{-1} \left[\frac{1}{(s-2)^2+3^2} \right]$$

$$e^{2t} \cos 3t + \frac{4}{3} e^{2t} \sin 3t$$

(ii) $\mathcal{L}^{-1} \left[\frac{s^2+6}{(s^2+1)(s^2+4)} \right]$

$$\frac{s^2+6}{(s^2+1)(s^2+4)} = \frac{As+B}{s^2+1} + \frac{Cs+D}{s^2+4}$$

$$s^2+6 = (As+B)(s^2+4) + (Cs+D)(s^2+1)$$

$$s^2+6 = As(s^2+4) + B(s^2+4) + Cs(s^2+1) + D(s^2+1)$$

$$s^3:- 0 = A+C \quad A=-C$$

$$s^2:- 1 = B+D \quad B=1-D$$

$$s:- 0 = 4A+C \quad \leftarrow$$

$$\text{const}:- 6 = 4B+D \quad \leftarrow$$

$$0 = -4C+C$$

$$\boxed{C=0} \quad \therefore \boxed{A=0}$$

$$6 = 4 - 4D + D$$

$$B = -3D$$

$$\boxed{D = -\frac{2}{3}}$$

$$\boxed{B = \frac{2}{3}}$$

$$B = 1 + \frac{2}{3} = \frac{5}{3}$$

$$\frac{s^2+6}{(s^2+1)(s^2+4)} = \frac{5}{3(s^2+1)} - \frac{2}{3(s^2+4)}$$

$$\mathcal{L}^{-1} \left[\frac{s^2+6}{(s^2+1)(s^2+4)} \right] = \frac{5}{3} \mathcal{L}^{-1} \left[\frac{1}{s^2+1} \right] - \frac{2}{3} \mathcal{L}^{-1} \left[\frac{1}{s^2+4} \right]$$

$$= \frac{5}{3} \sin t - \frac{1}{3} \sin 2t$$

$$(iii) \mathcal{L}^{-1} \left[\frac{s}{(s-3)(s^2+4)} \right]$$

$$\mathcal{L}^{-1} \left[\frac{s}{s^2+4} \cdot \frac{1}{s-3} \right]$$

$$\mathcal{L}^{-1} \left[\frac{s}{s^2+4} \right] = \cos 2t = f(t)$$

$$\mathcal{L}^{-1} \left[\frac{1}{s-3} \right] = e^{3t} = g(t)$$

~~$$\mathcal{L}^{-1} [f(t)g(t)] = \int_0^t f(u)g(t-u) du$$~~

$$\mathcal{L}^{-1} [F(s)G(s)] = \int_0^t f(u)g(t-u) du$$

$$= \int_0^t \cos 2u e^{3(t-u)} du$$

$$= e^{3t} \int_0^t e^{-3u} \cos 2u du$$

$$= e^{3t} \frac{e^{-3u}}{13} \left[-3 \cos 2u + 2 \sin 2u \right]_0^t$$

$$= \frac{e^{3t}}{13} \left[e^{-3t} (-3 \cos 2t + 2 \sin 2t) - e^0 (-3 \cos 0 + 2 \sin 0) \right]$$

$$= \frac{e^{3t}}{13} \left[e^{-3t} (-3 \cos 2t + 2 \sin 2t) + 3 \right]$$

$$\boxed{\frac{1}{13} [3e^{3t} - 3 \cos 2t + 2 \sin 2t]} \quad \underline{\underline{A}}$$

$$(iv) \mathcal{L}^{-1} \left[\frac{s^3}{s^4 - a^4} \right]$$

$$\mathcal{L}^{-1} \left[\frac{s^3}{(s^2-a^2)(s^2+a^2)} \right] = \mathcal{L}^{-1} \left[\frac{s^3}{(s^2-a^2)(s^2+a^2)} \right]$$

$$\frac{s^3}{(s^2-a^2)(s^2+a^2)} = \frac{As+B}{s^2-a^2} + \frac{Cs+D}{s^2+a^2}$$

$$s^3 = (As+B)(s^2+a^2) + (Cs+D)(s^2-a^2)$$

$$s^3:- A+C=1$$

$$s^2:- B+D=0$$

$$s:- Aa^2 - Ca^2 = 0$$

$$\text{const:- } Ba^2 - Da^2 = 0$$

$$\boxed{B=D}$$

$$2D=0 \quad \boxed{D=0}$$

$$\boxed{B=0}$$

$$A-C=0$$

$$A=C$$

$$\therefore C+C=1 \quad 2C=1 \quad \boxed{C=\frac{1}{2}}$$

$$\boxed{A=\frac{1}{2}}$$

$$\mathcal{L}^{-1} \left[\frac{s^3}{(s^2-a^2)(s^2+a^2)} \right] = \frac{1}{2} \mathcal{L}^{-1} \left[\frac{s}{s^2-a^2} \right] + \frac{1}{2} \mathcal{L}^{-1} \left[\frac{s}{s^2+a^2} \right]$$

$$\boxed{= \frac{1}{2} [\cosh(at) + \cos(at)]} \quad \underline{\underline{Ans}}$$

$$(v) \mathcal{L}^{-1} \left[\frac{S}{S^4 + 4a^4} \right]$$

$$\mathcal{L}^{-1} \left[\frac{S}{(S^2)^2 + (2a^2)^2 + 2(S^2)(2a^2) - 2(S^2)(2a^2)} \right]$$

$$\mathcal{L}^{-1} \left[\frac{S}{(S^2 + 2a^2)^2 - 4a^2 S^2} \right]$$

$$\mathcal{L}^{-1} \left[\frac{S}{(S^2 + 2a^2 + 2as)(S^2 + 2a^2 - 2as)} \right]$$

$$\frac{S}{(S^2 + 2as + 2a^2)(S^2 - 2as + 2a^2)} = \frac{As + B}{S^2 + 2as + 2a^2} + \frac{Cs + D}{S^2 - 2as + 2a^2}$$

$$S = (As + B)(S^2 - 2as + 2a^2) + (Cs + D)(S^2 + 2as + 2a^2)$$

$$S = As(S^2 - 2as + 2a^2) + B(S^2 - 2as + 2a^2) + Cs(S^2 + 2as + 2a^2) + D(S^2 + 2as + 2a^2)$$

$$\begin{aligned} \text{i) } S^3: & \quad A + C = 0 \quad \boxed{A = -C} \\ \text{ii) } S^2: & \quad -2aA + B + 2aC + D = 0 \quad \swarrow \left\{ \begin{aligned} 2aC + 2aC &= 0 \\ 4aC &= 0 \\ \boxed{C = 0} \end{aligned} \right. \\ \text{iii) } S: & \quad 2a^2A - 2aB + 2a^2C + 2aD = 1 \\ & \quad aA - B + aC + D = \frac{1}{2}a. \quad \leftarrow \begin{aligned} &\boxed{A = 0} \\ &\therefore \end{aligned} \\ \text{iv) Const:} & \quad -2a^2B + 2a^2D = 0 \\ & \quad B + D = 0 \quad \boxed{B = -D} \end{aligned}$$

for value of A & C in Qnd 4th

$$0 - B + 0 + D = \frac{1}{2}a.$$

$$-B + D = \frac{1}{2}a.$$

$$D + D = \frac{1}{2}a.$$

$$\boxed{D = \frac{1}{4}a}$$

$$\boxed{B = -\frac{1}{4}a}$$

$$\frac{s}{(s^2+2a^2+2as)(s^2+2a^2-2as)} = \frac{-1}{4a(s^2+2a^2+2as)} + \frac{1}{4a(s^2+2a^2-2as)}$$

$$\mathcal{L}^{-1}\left[\frac{s}{(s^2+2a^2+2as)(s^2+2a^2-2as)}\right] = \frac{-1}{4a} \mathcal{L}^{-1}\left[\frac{1}{s^2+2a^2+2as}\right] + \frac{1}{4a} \mathcal{L}^{-1}\left[\frac{1}{s^2+2a^2-2as}\right]$$

$$= \frac{-1}{4a} \mathcal{L}^{-1}\left[\frac{1}{(s+a)^2+a^2}\right] + \frac{1}{4a} \mathcal{L}^{-1}\left[\frac{1}{(s-a)^2+a^2}\right]$$

$$= \frac{-1}{4a} e^{-at} \cdot \frac{1}{a} \sin at + \frac{1}{4a} e^{at} \cdot \frac{1}{a} \sin at$$

$$= \frac{-1}{4a^2} e^{-at} \sin at + \frac{1}{4a^2} e^{at} \sin at$$

$$= + \frac{\sin at}{2a^2} \left[\frac{e^{at} - e^{-at}}{2} \right]$$

$$\left[\frac{e^x - e^{-x}}{2} = \sinh x \right]$$

$$= + \frac{\sin at}{2a^2} \cdot \sinh(at) = \boxed{\frac{1}{2a^2} \sin(at) \sinh(at)} \underline{\underline{Ans}}$$

$$(vi) \mathcal{L}^{-1}\left[\log \frac{s^2+1}{s(s+1)}\right] = f(t)$$

$$\mathcal{L}^{-1}\left[\frac{d}{ds} \log \frac{s^2+1}{s(s+1)}\right] = -tf(t)$$

$$\mathcal{L}^{-1}\left[\frac{d}{ds} [\log(s^2+1) - \log s(s+1)]\right] = -tf(t)$$

$$\mathcal{L}^{-1}\left[\frac{2s}{s^2+1} - \frac{2s+1}{s(s+1)}\right] = -tf(t)$$

$$\left. \begin{aligned} \mathcal{L}[t^n f(t)] &= (-1)^n \frac{d^n}{ds^n} [\phi(s)] \\ (-1)^n t^n f(t) &= \mathcal{L}^{-1}\left[\frac{d^n}{ds^n} \phi(s)\right] \end{aligned} \right\} \quad n=1$$

$$\{-tf(t) = \mathcal{L}^{-1}\left[\frac{d}{ds} \phi(s)\right]\}$$

$$\sin t - \cancel{\left[\frac{1}{s} - \frac{1}{s+1}\right]} = -tf(t)$$

$$\sin t - 1 + e^{-t} = -tf(t)$$

$$2\mathcal{L}^{-1}\left[\frac{s}{s^2+1}\right] - \left[\mathcal{L}^{-1}\left[\frac{1}{s}\right] + \mathcal{L}^{-1}\left[\frac{1}{s+1}\right]\right] = -tf(t)$$

$$2\cos t - 1 - e^{-t} = -tf(t)$$

$$f(t) = \boxed{\frac{1}{t}(1 + e^{-t} - 2\cos t)} \underline{\underline{Ans}}$$

$$(vii) \mathcal{L}^{-1} \left[\tan^{-1} \frac{2}{s^2} \right] = f(t)$$

$$\mathcal{L}^{-1} \left[\frac{d}{ds} \left(\tan^{-1} \frac{2}{s^2} \right) \right] = -t f(t)$$

$$\mathcal{L}^{-1} \left[\frac{1}{1 + \frac{4}{s^4}} \cdot \frac{2}{s^3} \right] = -t f(t)$$

$$-4 \mathcal{L}^{-1} \left[\frac{s^4}{s^4 + 4} \cdot \frac{1}{s^3} \right] = -t f(t)$$

$$-4 \mathcal{L}^{-1} \left[\frac{s}{(s^2)^2 + 2^2} \right] = -t f(t)$$

$$-4 \mathcal{L}^{-1} \left[\frac{s}{\underbrace{(s^2)^2 + 2^2 + 2(s^4)(2) - 2(s^4)(2)}} \right] = -t f(t)$$

$$-4 \mathcal{L}^{-1} \left[\frac{s}{(s^2+2)^2 - (2s)^2} \right] = -t f(t)$$

$$-4 \mathcal{L}^{-1} \left[\frac{s}{(s^2+2+2s)(s^2+2-2s)} \right] = -t f(t) \quad \text{--- (1)}$$

$$\frac{s}{(s^2+2s+2)(s^2-2s+2)} = \frac{(As+B)}{s^2+2s+2} + \frac{(Cs+D)}{s^2-2s+2}$$

$$S = As^3 + 2As^2 + 2As + Bs^2 + 2Bs + 2B + Cs^3 - 2Cs^2 + 2Cs + Ds^2 - 2Ds + 2D$$

$$s^3: - A + C = 0 \quad A = -C$$

$$s^2: - 2A + B - 2C + D = 0$$

$$s: 2A + 2B + 2C - 2D = 1$$

$$\text{Const: } 2B + 2D = 0$$

$$B + D = 0 \quad B = -D$$

$$0 - 2D + 0 - 2D = 1$$

$$-4D = 1$$

$$D = -\frac{1}{4}$$

$$\therefore B = \frac{1}{4}$$

$$\mathcal{L}^{-1}\left[\frac{s}{(s^2+2s+2)(s^2-2s+2)}\right] = \frac{1}{4}\mathcal{L}^{-1}\left[\frac{1}{s^2+2s+2}\right] - \frac{1}{4}\mathcal{L}^{-1}\left[\frac{1}{s^2-2s+2}\right]$$

$$= \frac{1}{4}\mathcal{L}^{-1}\left[\frac{1}{s^2-2(s)(1)+1+1}\right] - \frac{1}{4}\mathcal{L}^{-1}\left[\frac{1}{s^2-2(s)(1)+1+1}\right]$$

$$= \frac{1}{4}\mathcal{L}^{-1}\left[\frac{1}{(s-1)^2+1}\right] - \frac{1}{4}\mathcal{L}^{-1}\left[\frac{1}{(s+1)^2+1}\right]$$

$$= \frac{1}{4}e^t \sin t - \frac{1}{4}e^{-t} \sin t$$

$$\mathcal{L}\left[\frac{s}{(s^2+2s+2)(s^2-2s+2)}\right] = \frac{1}{2} \sin t \left[\frac{e^t - e^{-t}}{2} \right] = \boxed{\frac{1}{2} \sin t \sinh t}$$

for value in eqⁿ ①.

$$\mathcal{F}^{-1}\left[\frac{1}{2} \sin t \sinh t\right] = t f(t)$$

$$\boxed{\frac{2}{t} \sin t \sinh t = f(t)} \quad \underline{\underline{\text{Ans}}}$$

$$\text{(viii)} \quad \mathcal{L}^{-1}\left[\frac{s+2}{(s^2+4s+5)^2}\right]$$

$$\mathcal{L}^{-1}\left[\frac{1}{s^2+4s+5}\right] = \mathcal{L}^{-1}\left[\frac{1}{s^2+2(s)(2)+2^2-2^2+5}\right]$$

$$= \mathcal{L}^{-1}\left[\frac{1}{(s+2)^2+1}\right]$$

$$\mathcal{L}^{-1}\left[\frac{1}{s^2+4s+5}\right] = e^{-2t} \sin t$$

$$\mathcal{L}^{-1}\left[\frac{d}{ds}\left[\frac{1}{s^2+4s+5}\right]\right] = -t e^{-2t} \sin t$$

$$\mathcal{L}^{-1}\left[\frac{-(2s+4)}{(s^2+4s+5)^2}\right] = -t e^{-2t} \sin t$$

$$-2\mathcal{L}^{-1}\left[\frac{s+2}{(s^2+4s+5)^2}\right] = -t e^{-2t} \sin t$$

$$\mathcal{L}^{-1}\left[\frac{s+2}{(s^2+4s+5)^2}\right] = \boxed{\frac{t}{2} e^{-2t} \sin t} \quad \underline{\underline{\text{Ans}}}$$

⑦ Statement:- If $L^{-1}[F(s)] = f(t)$ and $L^{-1}[G(s)] = g(t)$

then $L^{-1}[F(s)G(s)] = \int_0^t f(u)g(t-u)du$.

Proof:- Let $\phi(t) = \int_0^t f(u)g(t-u)du$ — (1)

$$L[\phi(t)] = L\left[\int_0^t f(u)g(t-u)du\right]$$

$$L[\phi(t)] = \int_0^\infty \int_0^t e^{-st} f(u)g(t-u)du dt.$$

$$= \int_0^\infty \int_u^\infty e^{-st} f(u)g(t-u)dt du.$$

$$= \int_0^\infty \int_u^\infty e^{-st-su+su} f(u)g(t-u)dt du.$$

$$= \int_0^\infty e^{-su} f(u) \left[\int_u^\infty e^{-s(t-u)} g(t-u)dt \right] du.$$

$$= \int_0^\infty e^{-su} f(u) \left[\int_0^\infty e^{-sv} g(v)dv \right] du.$$

$$L[\phi(t)] = \int_0^\infty e^{-su} f(u) G(s) du.$$

$$= G(s) \int_0^\infty e^{-su} f(u) du.$$

$$L[\phi(t)] = G(s)F(s).$$

$$LHS = RHS$$

Hence proved.

$\Rightarrow L^{-1}\left[\frac{1}{(s^2+1)(s^2+9)}\right]$

$$L^{-1}\left[\frac{1}{s^2+1} \times \frac{1}{s^2+9}\right]$$

$$L^{-1}\left[\frac{1}{s^2+1}\right] = \sin t = f(t)$$

$$L^{-1}\left[\frac{1}{s^2+9}\right] = \frac{1}{3}\sin t = g(t)$$

$$\mathcal{L}^{-1} \left[\frac{1}{(s^2+1)(s^2+9)} \right] = \frac{1}{3} \int_0^t 2 \sin u \sin 3(t-u) du.$$

$$= \frac{1}{6} \int_0^t [\cos(u-3t+3u) - \cos(u+3t-3u)] du$$

$$= \frac{1}{6} \int_0^t [\cos[4u-3t] - \cos[3t-2u]] du.$$

$$= \frac{1}{6} \left[\frac{\sin(4u-3t)}{4} + \frac{\sin(3t-2u)}{2} \right]_0^t.$$

$$= \frac{1}{6} \left[\left(\frac{\sin t}{4} + \frac{\sin t}{2} \right) - \left(\frac{\sin(-3t)}{4} + \frac{\sin 3t}{2} \right) \right]$$

$$= \frac{1}{6} \left[\frac{3 \sin t}{4} - \underbrace{\left(-\frac{\sin 3t}{4} + \frac{\sin 3t}{2} \right)}_{\text{LCM}} \right]$$

$$= \frac{1}{6} \left[\frac{3 \sin t}{4} + \frac{\sin 3t}{4} \right] = \boxed{\frac{1}{24} [3 \sin t - \sin 3t]} \underline{\underline{Ans}}$$

⑧ $\mathcal{L}^{-1} \left[\frac{s}{(s^2+a^2)^2} \right]$ (Using Convolution Theorem)

$$\mathcal{L}^{-1} \left[\frac{s}{s^2+a^2} \cdot \frac{1}{s^2+a^2} \right]$$

$$\mathcal{L}^{-1} \left[\frac{s}{s^2+a^2} \right] = \cos at = f(t)$$

$$\mathcal{L}^{-1} \left[\frac{1}{s^2+a^2} \right] = \frac{\sin at}{a} = g(t)$$

$$\mathcal{L}^{-1} [F(s)G(s)] = \int_0^t f(u)g(t-u) du$$

$$= \frac{1}{a} \int_0^t \sin au \cos a(t-u) du$$

$$= \frac{1}{2a} \int_0^t 2 \sin au \cos (at-au) du$$

$$= \frac{1}{2a} \int_0^t [\sin (at+at-2au) + \sin (au-at+au)] du$$

$$= \frac{1}{2a} \int_0^t [\sin at + \sin (2au-at)] du$$

$$= \frac{1}{2a} \left[u \sin at - \frac{\cos (2au-at)}{2a} \right]_0^t$$

$$= \frac{1}{2a} \left[\left(t \sin at - \frac{1}{2a} \cos at \right) - \left(0 - \frac{\cos (-at)}{2a} \right) \right]$$

$$\cos(\theta) = \cos \theta$$

$$= \frac{1}{2a} \left[t \sin at - \frac{1}{2a} \cos at + \frac{1}{2a} \cos at \right]$$

$$\boxed{= \frac{1}{2a} t \sin at} \underline{\underline{\text{Ans}}}$$

(8) (Using derivative)

$$\mathcal{L}^{-1}\left[\frac{s}{(s^2+a^2)^2}\right]$$

$$\mathcal{L}^{-1}\left[\frac{1}{s^2+a^2}\right] = \frac{1}{a} \sin at$$

$$\mathcal{L}^{-1}\left[\frac{d}{ds}\left[\frac{1}{s^2+a^2}\right]\right] = -\frac{t}{a} \sin at$$

$$\mathcal{L}^{-1}\left[\frac{-2s}{(s^2+a^2)^2}\right] = -\frac{t}{a} \sin at$$

$$+2 \mathcal{L}^{-1}\left[\frac{s}{(s^2+a^2)^2}\right] = \frac{1}{a} t \sin at$$

$$\mathcal{L}^{-1}\left[\frac{s}{(s^2+a^2)^2}\right] = \boxed{\frac{1}{2a} t \sin at} \quad \underline{\underline{A}}$$

(9) Solve the following differential equations:-
(i) $y'' - 3y' + 2y = e^{3t}$ $y(0) = 1$ $y'(0) = 0$

$$\mathcal{L}[y''] - 3\mathcal{L}[y'] + 2\mathcal{L}[y] = \mathcal{L}[e^{3t}]$$

$$s^2 \mathcal{L}[y] - sy(0) - s^0 y'(0) - 3[s\mathcal{L}[y] - y(0)] + 2\mathcal{L}[y] = \frac{1}{s-3}$$

$$s^2 \mathcal{L}[y] - s - 3s\mathcal{L}[y] + 2\mathcal{L}[y] + 3 = \frac{1}{s-3}$$

$$\mathcal{L}[y][s^2 - 3s + 2] - s + 3 = \frac{1}{s-3}$$

$$\mathcal{L}[y] = \frac{1}{(s-3)(s^2-3s+2)} + \frac{s-3}{(s^2-3s+2)}$$

$$\mathcal{L}[y] = \frac{1}{(s-3)(s-1)(s-2)} + \frac{s-3}{(s-1)(s-2)}$$

$$\cancel{\mathcal{L}[y]} \quad y = \mathcal{L}^{-1}\left[\frac{1}{(s-1)(s-2)(s-3)}\right] + \mathcal{L}^{-1}\left[\frac{s-3}{(s-1)(s-2)}\right]$$

$$y = \mathcal{L}^{-1}\left[\frac{1}{2(s-1)} - \frac{1}{s-2} + \frac{1}{2(s-3)}\right] + \mathcal{L}^{-1}\left[\frac{2}{s-1} - \frac{1}{s-2}\right]$$

$$y = \frac{1}{2}e^t - e^{2t} + \frac{1}{2}e^{3t} + 2e^t - e^{2t} = \boxed{\frac{5}{2}e^t - 2e^{2t} + \frac{1}{2}e^{3t}} \quad \underline{\underline{A}}$$

$$(11) \frac{d^2x}{dt^2} + 9x = \cos 3t \quad \text{if } x(0) = 1, \quad x\left(\frac{\pi}{2}\right) = -1$$

$$x'' + 9x = \cos 3t$$

$$\mathcal{L}[x''] + 9\mathcal{L}[x] = \mathcal{L}[\cos 3t]$$

$$s^2 \mathcal{L}[x] - sx(0) - x'(0) + 9\mathcal{L}[x] = \frac{s}{s^2+9}$$

$$\mathcal{L}[x](s^2+9) - s - a = \frac{s}{s^2+9}$$

$$\mathcal{L}[x] = \frac{s}{(s^2+4)(s^2+9)} + \frac{s+a}{(s^2+9)}$$

$$x = \mathcal{L}^{-1}\left[\frac{s}{(s^2+4)(s^2+9)}\right] + \mathcal{L}^{-1}\left[\frac{s+a}{s^2+9}\right]$$

$$\frac{s}{(s^2+4)(s^2+9)} = \frac{As+B}{s^2+4} + \frac{Cs+D}{s^2+9}$$

$$s = (As+B)(s^2+9) + (Cs+D)(s^2+4)$$

$$\underline{s^3} :- A+C=0 \quad A=-C$$

$$\underline{s^2} :- B+D=0 \quad B=-D$$

$$\underline{s} :- 9A+4C=1 \quad \leftarrow \begin{array}{l} -9C+4C=1 \\ -5C=1 \end{array}$$

$$\underline{\text{const}} :- 9B+4D=0$$

$$-9D+4D=0$$

$$\boxed{D=0} \quad \boxed{B=0}$$

$$\boxed{C=-\frac{1}{5}}$$

$$\boxed{A=\frac{1}{5}}$$

$$\mathcal{L}^{-1}\left[\frac{s}{(s^2+4)(s^2+9)}\right] = \frac{1}{5}\mathcal{L}^{-1}\left[\frac{s}{s^2+4}\right] - \frac{1}{5}\mathcal{L}^{-1}\left[\frac{s}{s^2+9}\right]$$

$$= \frac{1}{5}\cos 2t - \frac{1}{5}\cos 3t$$

$$\mathcal{L}^{-1}\left[\frac{s+a}{s^2+9}\right] = \mathcal{L}^{-1}\left[\frac{s}{s^2+9}\right] + a\mathcal{L}^{-1}\left[\frac{1}{s^2+9}\right] = \cos 3t + \frac{a}{3}\sin 3t$$

$$\therefore x = \frac{1}{5}\cos 2t - \frac{1}{5}\cos 3t + \cos 3t + \frac{a}{3}\sin 3t$$

$$-1 = \frac{1}{5}\cos \pi + \frac{4}{5}\cos \frac{3\pi}{2} + \frac{a}{3}\sin \frac{3\pi}{2}$$

$$-1 = \frac{1}{5} + 0 - \frac{a}{3}$$

$$\boxed{a = 12/5}$$

$$x = \frac{1}{5}\cos 2t + \frac{4}{5}\cos 3t + \frac{4}{5}\sin 3t$$

$$x = \frac{1}{5}[\cos 2t + 4\sin 3t + 4\cos 3t]$$

Ans

$$(ii) \frac{d^4 y}{dt^4} - k^4 y = 0 \quad y(0) = 1 \quad y'(0) = y''(0) = y'''(0) = 0$$

$$\mathcal{L}[y^{(4)}] - k^4 \mathcal{L}[y] = 0$$

$$s^4 \mathcal{L}[y] - s^3 y(0) - s^2 y'(0) - s y''(0) - y'''(0) - k^4 \mathcal{L}[y] = 0$$

$$s^4 \mathcal{L}[y] - s^3 - k^4 \mathcal{L}[y] = 0$$

$$\mathcal{L}[y][s^4 - k^4] - s^3 = 0$$

$$\mathcal{L}[y] = \frac{s^3}{s^4 - k^4}$$

$$y = \mathcal{L}^{-1} \left[\frac{s^3}{s^4 - k^4} \right] = \mathcal{L}^{-1} \left[\frac{s^3}{(s^2 - k^2)(s^2 + k^2)} \right]$$

$$\frac{s^3}{(s^2 - k^2)(s^2 + k^2)} = \frac{As + B}{s^2 - k^2} + \frac{Cs + D}{s^2 + k^2}$$

$$\frac{s^3}{s^2} :- A + C = 1$$

$$\frac{s^2}{s^2} :- B + D = 0 \leftarrow 2D = 0 \quad (D = 0)$$

$$\frac{s}{s^2} :- Ak^2 - Ck^2 = 0$$

$$\text{Const} :- Bk^2 - Dk^2 = 0$$

$$B = D \quad (B = 0)$$

$$A - C = 0, A = C \quad (A = \frac{1}{2})$$

$$2C = 1 \quad (C = \frac{1}{2})$$

$$y = \mathcal{L}^{-1} \left[\frac{s^3}{(s^2 - k^2)(s^2 + k^2)} \right] = \frac{1}{2} \mathcal{L}^{-1} \left[\frac{s}{s^2 - k^2} \right] + \frac{1}{2} \mathcal{L}^{-1} \left[\frac{s}{s^2 + k^2} \right]$$

$$= \frac{1}{2} [\cosh(kt) + \cos(kt)] \quad \underline{\underline{A}}$$

(10) Second Shifting Theorem

$$\text{If } \mathcal{L}[f(t)] = F(s) \text{ \& } g(t) = \begin{cases} 0 & t < a \\ f(t-a) & t \geq a \end{cases}$$

$$\text{then, } \mathcal{L}[g(t)] = e^{-as} F(s) \text{ or}$$

$$\mathcal{L}[f(t-a)u(t-a)] = e^{-as} F(s)$$

Proof :- $\mathcal{L}[f(t-a)u(t-a)] = \int_0^\infty e^{-st} f(t-a)u(t-a) dt$

$$= \int_0^a e^{-st} (0) dt + \int_a^\infty e^{-st} f(t-a) dt$$

$$= \int_a^\infty e^{-st} f(t-a) dt$$

$$\begin{aligned} t-a &= v \\ t &= a+v \end{aligned}$$

$$\left. \begin{aligned} &\text{when } t=a \\ &\quad v=0 \\ &\text{when } t=\infty \\ &\quad v=\infty \end{aligned} \right\}$$

$$= \int_0^\infty e^{-s(a+v)} f(v) dv$$

$$= e^{-sa} \int_0^\infty e^{-sv} f(v) dv = e^{-sa} \mathcal{L}[f(v)]$$

$$\boxed{e^{-sa} F(s)} = \text{RHS}$$

Hence proved

(11) $f(t) = \begin{cases} \cos t & 0 < t < \pi \\ 1 & \pi < t < 2\pi \\ \sin t & t > 2\pi \end{cases}$

$$f(t) = \cos t [u(t-0) - u(t-\pi)] + 1 [u(t-\pi) - u(t-2\pi)] + \sin t [u(t-2\pi)]$$

$$\mathcal{L}[f(t)] = \mathcal{L}[u(t-0)\cos t] + \mathcal{L}[u(t-\pi)[1-\cos t]] + \mathcal{L}[u(t-2\pi)(\sin t-1)]$$

$$= \mathcal{L}[u(t-0)\cos(t-0)] + \mathcal{L}[u(t-\pi)(1-\cos(t-\pi))] + \mathcal{L}[u(t-2\pi)(\sin(t-2\pi)-1)]$$

$$= e^{-0s} \left(\frac{s}{s^2+1} \right) + e^{-\pi s} \left(\frac{1}{s} + \frac{s}{s^2+1} \right) + e^{-2\pi s} \left(\frac{1}{s^2+1} - \frac{1}{s} \right) \quad \underline{\underline{Ans}}$$

(12) Find inverse Laplace transform:-

(i) $\mathcal{L}^{-1} \left[\frac{e^{-2s}}{s^2+8s+25} \right]$

$$\mathcal{L}^{-1} \left[\frac{1}{s^2+8s+25} \right] = \mathcal{L}^{-1} \left[\frac{1}{s^2+2(s)(4)+4^2-4^2+25} \right]$$

$$\mathcal{L}^{-1} \left[\frac{1}{(s+4)^2+3^2} \right] = \boxed{\frac{e^{-4t}}{3} \sin 3t} \rightarrow f(t)$$

$$\mathcal{L}^{-1}[e^{-as} F(s)] = f(t-a) u(t-a)$$

here $a=2$

$$= \frac{e^{-4(t-2)}}{3} \sin 3(t-2) = \boxed{\frac{1}{3} e^{-4(t-2)} \sin 3(t-2)} \underline{\underline{Ans}}$$

$$(ii) \mathcal{L}^{-1}\left[\frac{e^{-s}}{(s+1)^3}\right]$$

$$\mathcal{L}^{-1}\left[\frac{1}{(s+1)^3}\right] = \frac{e^{-t} t^2}{2!}$$

$$\mathcal{L}^{-1}\left[\frac{1}{(s+1)^3}\right] = \frac{e^{-t} \cdot t^2}{2}$$

$$\left[\begin{array}{l} \mathcal{L}[t^n] = \frac{n!}{s^{n+1}} \\ \mathcal{L}[t^2] = \frac{2!}{s^{2+1}} \\ \frac{t^2}{2!} = \mathcal{L}^{-1}\left[\frac{1}{s^3}\right] \end{array} \right]$$

$$\mathcal{L}^{-1}[e^{-as} F(s)] = f(t-a) u(t-a)$$

here $a=1$

$$\mathcal{L}^{-1}\left[e^{-s} \cdot \frac{1}{(s+1)^3}\right] = \boxed{\frac{1}{2} e^{-(t-1)} (t-1)^2 u(t-1)} \underline{\underline{Ans}}$$

$$(iii) \mathcal{L}^{-1}\left[\frac{3}{s} - 4 \frac{e^{-s}}{s^2} + \frac{4e^{-3s}}{s^2}\right]$$

$$\boxed{\underbrace{3 \mathcal{L}^{-1}\left[\frac{1}{s}\right]}_{\text{I}} - \underbrace{4 \mathcal{L}^{-1}\left[\frac{e^{-s}}{s^2}\right]}_{\text{II}} + \underbrace{4 \mathcal{L}^{-1}\left[\frac{e^{-3s}}{s^2}\right]}_{\text{III}}} \quad \text{--- (1)}$$

$$1) \mathcal{L}^{-1}\left[\frac{1}{s}\right] = \boxed{1}$$

$$2) \mathcal{L}^{-1}\left[\frac{e^{-s}}{s^2}\right] = ?$$

$$\mathcal{L}^{-1}\left[\frac{1}{s^2}\right] = t$$

$$\mathcal{L}^{-1}\left[\frac{1}{s^{n+1}}\right] = \frac{t^n}{n!}$$

$$\mathcal{L}^{-1}[e^{-as} F(s)] = f(t-a) u(t-a)$$

here $a=1$

$$\mathcal{L}^{-1}\left[e^{-s} \cdot \frac{1}{s^2}\right] = \boxed{(t-1) u(t-1)}$$

$$\text{III) } \mathcal{L}^{-1}\left[\frac{e^{-3s}}{s^2}\right]$$

$$\mathcal{L}^{-1}\left[\frac{1}{s^2}\right] = t.$$

$$\mathcal{L}^{-1}[e^{-as} F(s)] = f(t-a) U(t-a)$$

here $a = 3$.

$$\mathcal{L}^{-1}\left[e^{-3s} \frac{1}{s^2}\right] = (t-3) U(t-3)$$

Put all values in eqⁿ (1)

$$3(1) - 4(t-1)U(t-1) + 4(t-3)U(t-3) \quad \underline{\underline{A}}$$